

# Yutong (Jennifer) Zhong

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## EDUCATION

### New York University

New York, USA

Bachelor of Arts in Computer Science, Minor in Mathematics, College of Arts and Science

Sep. 2023-May. 2027

GPA: 3.85/4.0 | Dean's List for Academic Year, 2023–2024 and 2024–2025 | Fundamentals of Machine Learning; Natural Language Processing; Data Management and Analysis; Computer Vision

## RESEARCH PROJECTS

### New York University--Undergraduate Research in Computer Science: Reliable RAG for Spreadsheet QA

New York, USA

Researcher

Sep. 2025 – Jun. 2026

- Conducted undergraduate honors research on reliability failures in RAG-based question answering, focusing on retrieval incompleteness, metric ambiguity, and hidden assumptions in spreadsheet aggregation tasks.
- Built a hybrid SQL–RAG pipeline that represents spreadsheet data in both a structured SQL database and a semantic index, using retrieval for natural-language evidence matching and SQL verification for exact filtering, grouping, ranking, and aggregation.
- Designed and evaluated ambiguity-aware response strategies, including direct answering, clarification-first answering, and dual-output multi-interpretation answering, to reduce silent metric-assumption errors in business QA.
- Evaluated the system on 25 ambiguous business questions with 75 total evaluations, measuring silent error rate, ambiguity detection correctness, and interpretation coverage; found that clarification-first answering eliminated silent metric-assumption errors, while dual-output improved ambiguity detection but still suffered from partial coverage.

### University of Liverpool--T-MOTR: Temporal-enhanced MOTR for Multi-Object Tracking on DanceTrack

Shanghai, China

Remote Research Assistant

Aug. 2025 – Nov. 2025

- Developed T-MOTR, an enhanced end-to-end Transformer-based multi-object tracker built on top of MOTR, targeting performance challenges in the DanceTrack benchmark characterized by uniform appearance and diverse motion.
- Proposed and implemented a cross-attention gating module to strengthen inter-frame query association and suppress redundant or noisy target activations.
- Integrated a temporal self-attention mechanism across decoder outputs to improve trajectory coherence under motion blur and occlusion conditions.
- Designed a temporal consistency loss combining inter-frame IoU and trajectory smoothness, improving HOTA by 3.1 points and IDF1 by 4.7 points on the DanceTrack validation set over the original MOTR baseline.

### Central South University--Efficient Multi-Modal Behavior Understanding

Hunan, China

Research Assistant

May. 2025 – Aug. 2025

- Identified a 2D semantic shortcut in multimodal 3D grounding using 2D-only ScanRefer diagnostics and t-SNE feature visualization, showing that fused representations rely heavily on 2D semantic cues while underusing reconstructed 3D geometry.
- Proposed W2R2, a What–Where Representation Reforming framework with two shared-parameter forward passes: a full 2D+3D fused path for alignment and a 2D-only shortcut path for conditional shortcut deterrence.
- Designed a pull–push objective combining standard alignment loss with a shortcut deterrence loss, encouraging 2D features to retain semantic “what” cues while making 3D features dominate spatial “where” grounding; evaluated on ScanRefer, Scan2Cap, and ScanQA without changing the inference-time architecture.

### University of Liverpool--LipSwin: Lightweight Swin Transformer for Visual Speech Recognition

Shanghai, China

Remote Research Assistant

Dec. 2024 – May. 2025

- Implemented LipSwin, a Swin Transformer–based visual encoder for word- and sentence-level lip-reading on LRW, LRS2, and LRS3.
- Designed a 3D spatio-temporal embedding module to preserve fine-grained lip-motion dynamics by avoiding early spatial pooling and retaining richer visual context.
- Incorporated light-weight 1D Convolutional Attention Module (Conformer-inspired) to improve temporal feature extraction and sequence modeling, while reducing attention cost.

### High School Research in RSA and OAEP Padding

Shenzhen, China

Independent Researcher

Jun. 2021 – Sept. 2021

- Reviewed RSA public-key cryptography, textbook RSA vulnerabilities, and OAEP padding as a randomized scheme for improving semantic security.
- Summarized chosen-plaintext and blinding attacks on deterministic RSA and explained how OAEP mitigates information leakage through randomized encoding.

## INTERNSHIP, TEACHING & LEADERSHIP EXPERIENCE

### SIAT Shenzhen Institute of Advanced Integration Technology

Shenzhen, China

Research Assistant

May. 2026 – Aug. 2026

- Conducted research and development on maritime intelligent surveillance systems for small-target detection and early warning using multi-source sensing data, including video streams and radar signals.
- Built and iteratively improved deep learning pipelines for maritime target recognition, tracking, and abnormal-behavior warning, covering data preprocessing, model training, inference, and deployment-oriented evaluation.
- Participated in multi-modal data collection and annotation workflows by integrating radar observations with video-based target information to improve target association and monitoring reliability in complex maritime environments.
- Developed experimental pipelines to evaluate detection robustness under challenging conditions such as long distance, occlusion, low visibility, and sparse target appearance; analyzed failure cases and supported system-level warning logic design.

### PricewaterhouseCoopers

Shenzhen, China

Data Analyst internship

Jun. 2024 – Aug. 2024

- Conducted data cleaning, analysis, and visualization using Python and Excel. Automated data workflows and created structured data tables to improve efficiency.

- Demonstrated initiative and strong collaboration by actively communicating with teammates and proposing data-driven solutions.

## **New York University**

Math Grader in Calculus I & Discrete Mathematics

NY, USA  
Sep. 2024 - *Present*

- Worked closely with professors and students to review and clarify advanced math homework; took initiative to organize peer discussions and coordinate group problem-solving sessions.

## **TANDON CSSA PM Dept.**

Organizer

NY, USA  
May. 2024 - May. 2025

- Led a small team to coordinate outreach efforts for student events; designed promotional content using tools like Xiumi and Canva, and developed strategies for engagement via social media.

## **PUBLICATIONS**

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- Zhong, Y. (2026). Where, Not What: Compelling Video LLMs to Learn Geometric Causality for 3D-Grounding. Accepted to ICASSP 2026. IEEE International Conference on Acoustics, Speech, and Signal Processing, Barcelona, Spain.
- Zhong, Y. (2022). An Overview of RSA and OAEP Padding. Highlights in Science, Engineering and Technology, 1, 82–86. <https://doi.org/10.54097/hset.v1i.431>

## **SKILLS AND AWARDS**

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- Skills: Python, Java, C, SQL, PyTorch, NumPy, Pandas and LaTeX.
- Research Experience: RAG pipelines, SQL-based question answering, computer vision model evaluation, and ablation analysis.
- Awards: Dean's Undergraduate Research Fund (DURF) Research Grant, CEMC Hypatia Contest Distinction (top 25%), Josh Goldfaden Award (fall 2023), HEIghen Critical Thinking Challenge Distinction. The Affiliated International School of Shenzhen University top 5% GPA (3.97).